Q1. What is a Function? State its types and explain the syntax to define a Function.

Ans:

A function is a block of reusable code that performs a specific task. We use functions to avoid writing the same code again and again. A function runs when it is called.

Types of Functions in Python:

There are mainly two types of functions:

1. Built-in Functions: These are already available in Python. We can directly use them.

Examples: print(), len(), type(), input(), sum().

2. User-defined Functions: These are created by programmers using the def keyword to perform a specific task.

Example:

```
def greet():
    print("Hello")


greet()
```

Syntax to define a Function:

```
def function_name(parameters):
    # statements
    return value
```

Explanation:

- def – Keyword used to define a function.
- function_name – Name given to the function.
- parameters – Values accepted by the function (optional).
- : – Indicates the beginning of the function body.
- statements – Code executed inside the function.
- return – Sends the result back (optional).

Example:

```
def add(a, b):
    return a + b
print(add(10, 20))
```

Output:

30

Q2. What is a Lambda Function?

Ans:

A lambda function is a small anonymous function (a function without a name) in Python. It is defined using the lambda keyword and can contain only one expression. It is mainly used for short and simple operations.

Syntax:

lambda arguments: expression

Example:

```
add = lambda a, b: a + b


print(add(10, 20))
```

Output:

30

Explanation:

- lambda – Keyword used to create a lambda function.
- a, b – Arguments passed to the function.
- a + b – Expression that returns the result.
- add – Variable used to store the lambda function.